

What Code-Mixing Actually Breaks in Clinical Retrieval-Augmented Generation: A Measurement Study on Hinglish Patient Queries

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Abstract

Clinical decision-support systems assume queries arrive in clean formal English. In India they frequently arrive in Hinglish, a romanised Hindi-English code-switched register. We study a retrieval-augmented generation (RAG) pipeline for Hinglish patient questions grounded in English clinical case reports, and report a measurement study rather than a system-improvement result. Three findings follow. First, code-mixing damages retrieval, and it damages *lexical* retrieval roughly four times more than dense cross-lingual retrieval: on 3,015 query-case pairs, BM25 loses **0.0912** Recall@1 relative to a gold English rendering of the same question ($p = 9.9 \times 10^{-26}$) while a passage-chunked LaBSE index loses **0.0206** ($p = 0.017$). Second, the failure of an Indian-language-specialised encoder is a *script* rather than a language effect: MuRIL performs at the random floor on romanised Hinglish (**0.0640** versus **0.0626**) yet recovers to **0.1821** on the same content in English. Third, the apparent factuality benefit of grounding is largely a property of the evaluation protocol: scoring both arms against the retrieved evidence — text the grounded arm was conditioned on and the zero-shot arm never saw — yields **+0.203** concept F1, whereas scoring against a reference neither arm saw yields **+0.062** on one generator and **−0.047** on another. We further show that the concept-overlap metric in common use is precision-only, so a constant one-word answer outscores the real system by a factor of 4.7, and we release degenerate baselines that expose this. We argue that measurement design, not architecture, currently dominates reported RAG gains for code-mixed clinical text.

Keywords: code-switching, retrieval-augmented generation, clinical NLP, Hinglish, evaluation methodology, cross-lingual information retrieval

1 Introduction

Clinical natural-language systems are typically built and benchmarked on formal English. Real patient communication in India is frequently *code-switched*: a single question mixes Hindi and English, written in Latin script, with no standardised orthography. A patient may write “*Doctor, meri beti ko skin pe rash hai aur bahut khujli ho rahi hai*” where a report would say “*erythematous maculopapular rash with pruritus*”. The clinical content is present; the surface form is not what the system expects.

Retrieval-augmented generation (RAG) is an appealing response. If a system can retrieve an authoritative case report and condition its answer on that evidence, it should produce grounded explanations without hallucinating. This paper examines whether that promise holds for Hinglish, and finds that the answer depends less on the architecture than on how the evaluation is constructed.

We build a retrieve-then-generate pipeline over 3,015 Hinglish patient queries paired with English clinical case narratives, and subject it to a deliberately adversarial measurement study. Our contributions are:

- **A retrieval-stage code-mixing penalty, measured against gold human translations with an explicit leakage gate**, showing that lexical retrieval is approximately $4.4\times$ more damaged by code-mixing than dense cross-lingual retrieval (Section 5.1).
- **Evidence that script, not language, is the failure mode** for Indian-language-specialised encoders on romanised text (Section 5.1).
- **A demonstration that the standard concept-overlap protocol overstates grounding benefit**, together with the degenerate baselines and the unbiased reference needed to detect it (Sections 5.2–5.3).
- **Two negative results reported in full**: a published adaptive-truncation heuristic that does not transfer to this setting, and a configuration defect that silently discarded 85% of every indexed document (Section 5.4).

We report several results that are unfavourable to our own system. We do so deliberately: each was found by instrumenting the evaluation rather than the model, and each would otherwise have been discovered by a reviewer.

2 Related Work and Research Gap

Work on code-mixed clinical text has focused principally on *generation* — summarising or answering code-mixed medical questions — rather than on the retrieval stage that precedes it. Multimodal code-mixed question summarisation datasets provide Hinglish patient queries with English summaries, and RAG systems for medical vision–language models propose domain-aware retrieval with adaptive context selection. Evaluation in this literature typically relies on lexical overlap against a reference, or on concept-level factuality scores computed against retrieved evidence.

Two gaps motivate this study. First, the retrieval stage is rarely evaluated separately for code-mixed queries, so it is not known how much of the end-to-end degradation is attributable to retrieval rather than generation. Second, factuality metrics for code-mixed clinical generation are largely unvalidated: they are adopted

from English pipelines and applied without checking whether they discriminate good answers from degenerate ones. This paper addresses both.

3 Materials and Methods

3.1 Problem Formulation

Let q be a Hinglish patient query and $\mathcal{C} = \{c_1, \dots, c_N\}$ a corpus of English clinical case narratives. A retriever R scores each case and returns the top- k set $\mathcal{E}_k(q) \subset \mathcal{C}$. A generator G produces an answer $a = G(q, \mathcal{E}_k(q))$ conditioned on that evidence, against a zero-shot baseline $a_0 = G(q)$.

Retrieval is evaluated with a condition-group relevance criterion: a retrieved case is relevant if its condition group is among those assigned to the query. Generation is evaluated by clinical-concept overlap between an answer and a reference text r :

$$P = \frac{|\mathcal{K}(a) \cap \mathcal{K}(r)|}{|\mathcal{K}(a)|}, \quad R_c = \frac{|\mathcal{K}(a) \cap \mathcal{K}(r)|}{|\mathcal{K}(r)|}, \quad F_1 = \frac{2PR_c}{P + R_c} \quad (1)$$

where $\mathcal{K}(\cdot)$ extracts positively-asserted clinical concepts. The choice of r is the central methodological variable of this paper and is treated explicitly in Section 4.2.

3.2 Architecture

The pipeline encodes q with LaBSE, searches a FAISS inner-product index over L2-normalised passage embeddings, max-pools passage scores to case scores, and injects the retrieved case text into a grounded prompt. We compare four retrievers: dense (LaBSE over passages), BM25, TF-IDF, and a reciprocal-rank fusion (RRF) of dense and lexical rankings.

Passage chunking.

Encoding a case as a single vector is lossy: the median case is 554 words, while the encoder’s sequence limit admits roughly 170. We therefore split each case into overlapping 256-token windows (32-token overlap, at most six per case, 4.17 on average) and score a case by its best-matching window. This is what allows every retriever to read the same content, which the comparison in Section 5.1 requires.

Fusion depth.

RRF assigns $\text{score}(c) = \sum_i 1/(K + \text{rank}_i(c))$ with $K = 60$. Candidate lists are taken to depth 100 before fusion and truncated to k afterwards. This is a correctness requirement rather than a tuning choice: fusing two depth-10 lists cannot surface any document outside their union, and reduces the hybrid to a re-ordering of what both components already agreed on.

4 Experimental Design and Evaluation

4.1 Datasets and Preprocessing

Queries. MMCQSD supplies Hinglish patient questions, each with a human-written English summary and an image caption. **Evidence.** MultiCaRe supplies English clinical case reports; 61,316 cases were filtered and 10,000 indexed, balanced across 18 condition groups.

Constructing query–evidence pairs was itself a methodological finding. Pairing Open-i radiology reports to MMCQSD queries by TF-IDF yielded only 11 usable pairs, as the two corpora had almost no topical overlap. Replacing the corpus with MultiCaRe and the matcher with LaBSE plus condition-aware filtering produced **3,015 pairs at 100% query coverage**, a $274\times$ increase.

Leakage gate.

MMCQSD’s English summary contains an image caption that names the condition group verbatim in 96.2% of rows — that is, it contains the relevance label itself. Any “English versus Hinglish” comparison using the full summary is therefore inflated. We strip the caption and assert that no condition label survives into the English query condition. Table 1 reports the stripped condition only.

4.2 Baselines and Evaluation Metrics

Retrieval is reported as Recall@ k and MRR@10 against a prevalence-weighted random floor of 0.0626. Generation is reported as concept precision, recall and F_1 (Eq. 1) against *two* references:

- **Circular reference** — the retrieved case. The grounded arm was conditioned on this text; the zero-shot arm never saw it.
- **Unbiased reference** — the MMCQSD image description, a human-written account of clinical findings that *neither* arm saw.

Degenerate baselines.

Concept precision has no recall term, so its optimum is to assert a single common concept. We therefore report constant-answer baselines alongside every system. Their necessity is demonstrated in Section 5.2.

Statistics.

Paired comparisons use the Wilcoxon signed-rank test; binary retrieval outcomes use McNemar’s test; intervals are 10,000-sample bootstraps. Because the image description repeats across rows (412 distinct strings over 2,988 rows, one covering 22%), intervals on that reference resample *clusters* rather than rows. Benjamini–Hochberg correction is applied across the full family of generation contrasts.

4.3 Implementation Details

Encoder: `sentence-transformers/LaBSE`, 768-d, CPU, `max_seq_length` 256. Index: FAISS `IndexFlatIP`, 41,746 passages over 10,000 cases. Generator: Groq `openai/gpt-oss-120b`, temperature 0.3, `max_tokens` 300, evidence truncated to 400 words.

A reproducibility limitation.

The generator used for our largest generation run, `llama-3.1-8b-instant`, was decommissioned by the provider during this study and now returns HTTP 404. Those outputs are archived and re-scorable but cannot be regenerated. We therefore replicate on `gpt-oss-120b` and report both. We note this as a broader hazard for work built on hosted free-tier models.

5 Results

5.1 Retrieval and the code-mixing penalty

Table 1 reports Recall@1 for all four retrievers under both query conditions, with every system reading the same full case text.

Table 1 Retrieval quality by system and query language ($n = 3,015$). The penalty is Recall@1 under the English question minus Recall@1 under the Hinglish query.

System	Hinglish	English	Penalty	McNemar p
Hybrid (RRF)	0.1751	0.1973	+0.0222	0.018
LaBSE (passages)	0.1280	0.1486	+0.0206	0.017
BM25	0.0935	0.1847	+0.0912	9.9×10^{-26}
TF-IDF	0.0842	0.1529	—	—
Random floor	0.0626	0.0626	—	—

Two results follow. First, **the code-mixing penalty is significant for every retrieval method**, which is why we test it per system: a penalty that appears under one configuration is a property of that configuration, whereas one that holds across methods is a property of code-mixing. Second, **the penalty is 4.4× larger for lexical retrieval than for dense**. BM25 is the strongest system on English and the weakest on Hinglish; the dense retriever barely moves. Figure 1 shows this asymmetry with bootstrap intervals.

Script, not language.

On the single-vector index, MuRIL — pretrained on Indian languages — performs at the random floor for Hinglish (0.0640 versus a floor of 0.0626) but recovers to 0.1821 on the same content rendered in English. MuRIL is trained on Devanagari Hindi whereas

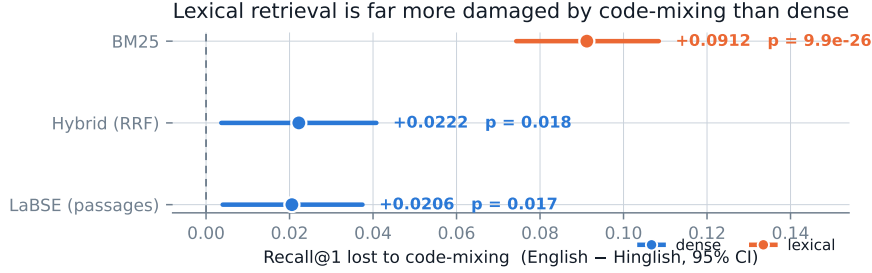


Fig. 1 Recall@1 lost to code-mixing, by retrieval system, with 95% bootstrap intervals. Lexical retrieval is far more damaged than dense.

MMCQSD is romanised. The failure is therefore one of *script* rather than of language, which has direct consequences for encoder selection in code-mixed settings.

5.2 The metric does not discriminate

Scored against the unbiased reference, a constant answer consisting of the single word “*swelling*” attains a concept precision of 0.7132. The grounded system attains 0.1528. A hard-coded word outperforms the real system by a factor of 4.7.

The cause is structural: precision alone rewards asserting few concepts. Absolute levels of this metric therefore carry no information about answer quality, and only paired differences between arms scored identically are interpretable. We report F_1 throughout and ship the constant-answer rows with every table. We further note that the commonly-reported “hallucination rate” is, under this definition, exactly $1 - \text{precision}$, so reporting both double-counts a single result.

5.3 Grounding depends on the reference

Table 2 scores the *same* generations against both references.

Table 2 Grounding effect on concept F_1 (grounded minus zero-shot), by reference. Benjamini–Hochberg corrected.

Generator	Evidence	Circular ref.	Unbiased ref.
llama-3.1-8b	oracle	+0.203	+0.062
gpt-oss-120b	oracle	+0.093	−0.021 (n.s.)
gpt-oss-120b	real retrieval	−0.032 (n.s.)	− 0.047

Under the circular reference, grounding appears strongly beneficial. Under a reference neither arm saw, the effect shrinks by a factor of three on one generator and becomes significantly *negative* on the other. The mechanism is visible in the components: grounding raises precision and lowers recall (oracle arm, unbiased reference: −0.1384 recall, BH $p = 0.00025$). Grounding makes the model more conservative,

and whether that trade is profitable depends on the generator. Figure 2 shows all six contrasts with intervals.

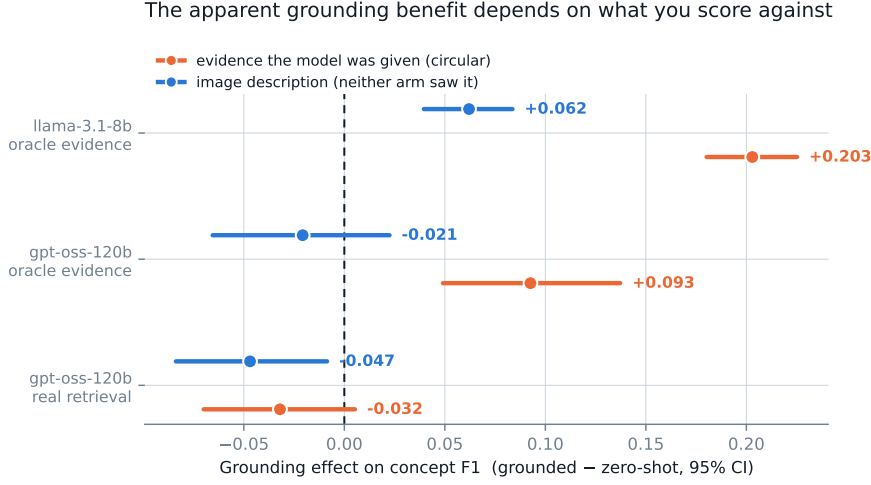


Fig. 2 The same generations scored against two references give opposite answers.

Refusal is a first-class outcome.

With real retrieval the grounded arm declines to answer in 33.8% of cases (zero-shot: 0.2%), typically stating that the evidence concerns a different patient. Refusals assert no clinical concept and therefore score as missing, disappearing from a naive mean — which would make the system appear healthiest exactly where it fails. We report answer coverage (49–52% for grounded arms, 92.9% zero-shot) alongside every factuality figure.

5.4 Code-mixing intensity, and two negative results

H₂: intensity.

Using a repaired code-mixing measure, grounded factual support is flat with respect to code-mixing intensity ($\rho = -0.0006$, $p = 0.98$) while the zero-shot arm degrades significantly ($\rho = -0.116$, BH $p = 0.0003$). Grounding absorbs the degradation that code-mixing induces in the ungrounded model. We note that the measure required repair: the original lexicon counted the English words *doctor* and *please* as Hindi, and they occur in 68.2% and 35.7% of queries respectively.

Adaptive truncation does not transfer.

A published similarity-gap heuristic for adaptive context selection fires on 0 of 3,015 queries at its specified threshold: it requires a 0.248 similarity gap between adjacent neighbours, and the largest gap occurring anywhere in the data is 0.109. Swept across thresholds and compared against a fixed k at matched evidence budget, it wins 0 of 6

settings. Precision is flat (0.112–0.115) at every setting while recall falls monotonically, indicating that the similarity gap carries no relevance information here.

A configuration defect dominated the retrieval comparison.

The index was originally built with a sequence limit of 128 tokens, truncating 100% of case narratives (median 307 tokens) and 52.5% of queries, while lexical baselines read the full document. Correcting this and adding passage chunking moved dense Recall@1 on Hinglish from 0.1144 to 0.1280 and reversed the ordering against BM25. We report this because the uncorrected comparison would have supported the opposite conclusion.

6 Discussion

The results support a single claim: for code-mixed clinical RAG, *measurement design currently dominates architecture* in determining reported gains. The same generations yield +0.203 or −0.047 concept F_1 depending only on the reference text; a one-word constant answer outscores the system by 4.7× on the standard metric; and the most-cited comparison in our own Table 1 reversed once a sequence-limit defect was corrected.

The retrieval findings are more robust and more actionable. The 4.4× asymmetry between lexical and dense degradation is a direct empirical argument for cross-lingual embedding in code-mixed settings, and the MuRIL result refines it: what matters is script coverage, not nominal language coverage. Practitioners deploying Indian-language encoders on romanised user text should expect near-chance retrieval.

6.1 Limitations and Threats to Validity

- **Relevance labels are coarse.** Condition-group matching over 18 groups admits cases that are same-group but clinically irrelevant. Absolute Recall values should be read as a lower bound.
- **The concept lexicon is not clinician-validated** and covers 26 concepts. It cannot capture findings outside that vocabulary.
- **The unbiased reference is narrow and repetitive.** The image description averages 1.5 concepts and takes 412 distinct values over 2,988 rows; we address the non-independence with cluster bootstraps but the construct remains partial.
- **Generator dependence.** Two of our generation results differ in sign between generators. We report both rather than selecting one.
- **Provider instability.** A model central to the study was withdrawn during it.
- **H₃ was not run.** Evidence-provenance comparison was cut: PubMedQA is only 2.1% on-topic for these conditions and MMedBench is 57.6% non-English, so the comparison would have measured corpus topicality rather than provenance.
- **No clinical deployment claim is made.** Absolute Recall@1 of 0.175 is far below what patient-facing use would require.

7 Conclusion and Future Work

We measured a Hinglish clinical RAG pipeline and found that code-mixing imposes a significant retrieval penalty which is four times larger for lexical than for dense retrieval; that an Indian-language encoder fails on romanised text for reasons of script rather than language; and that the apparent factuality benefit of grounding is substantially an artefact of scoring answers against the evidence they were conditioned on. We release the corrected evaluation, the degenerate baselines that expose the metric’s failure mode, and two negative results.

Future work should establish a random-evidence control to separate genuine grounding from evidence echoing, validate the concept lexicon against clinician annotation, and extend the analysis to the multimodal signal that MMCQSD provides but that this text-only study does not use.

Supplementary Information.

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Materials availability Not applicable.

Code availability All source code, prompts, configuration and analysis scripts are available at A single entry point regenerates every reported number and figure from cached artefacts.

Author contributions

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